Open-Core Architecture

AZRAS

Adaptive Zero-Rebuild Asset System

旧名称：RC アップライト工法 (RC Upright Method)

Technical White Paper v2.0

A Future-Oriented Renewable Architectural Platform

未来志向の再生可能建築プラットフォーム

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1. Executive Summary

AZRAS is a groundbreaking hybrid architectural system that dramatically minimizes the need for full rebuilding by separating a minimal, long-life Reinforced Concrete (RC) structural core from fully replaceable timber infill units.

First developed and physically verified in Japan in 2005, AZRAS surpasses conventional Skeleton-Infill (SI) systems through two decisive innovations:

- Complete Façade Freedom: No RC beams cross the façade plane, allowing each unit's exterior to be completely redesigned at every renewal cycle.
- Full Second-Floor Reconfigurability: The entire upper floor structure is timber, enabling complete changes to floor openings, voids, staircases, split-levels, and spatial layouts.

These capabilities allow not only interior renewal but also fundamental transformation of appearance, spatial quality, and functionality — while keeping the primary structural core intact over a very long period.

AZRAS delivers RC-level durability and seismic performance at approximately the same initial cost as light-gauge steel framing, while offering superior disaster resilience and exceptional compatibility with robotic construction and AGI/ASI-era technologies.

2. Development Background and Track Record

2.1 Origin

AZRAS was conceived around 2005 when a client sought rental housing that could simultaneously achieve low construction cost, high rental income, and long-term durability. The solution was to separate the structural lifespan (100+ years) from the market lifespan (20–30 years).

2.2 Verified Performance

Item	Detail
Development & Construction	2005
Buildings Completed	3 buildings (Japan)
Seismic Analysis	Dynamic nonlinear response analysis (2005)

On-site Verification	Full-scale vibration test by VIIC (2005)
Initial Cost	Equivalent to light-gauge steel framing
Achieved Rent	¥110,000 per unit (approx. 83% higher than ¥60,000 of comparable units)
Total Monthly Revenue	¥330,000 vs ¥240,000 (37.5% increase)

2.3 Current Status in Japan

After the 2005 projects, Japan significantly tightened structural review standards for hybrid RC-timber systems. This constraint is Japan-specific. Internationally, the system can be adapted to local codes and standards.

3. How AZRAS Differs from Conventional SI Systems

3.1 Limitations of Standard SI Systems

Conventional Skeleton-Infill systems allow interior reconfiguration but are constrained by RC moment frames, resulting in fixed façade lines and fixed second-floor RC slabs.

3.2 AZRAS's Two Fundamental Advantages

Complete Façade Freedom

One-directional RC load-bearing walls eliminate RC beams on the façade plane. The entire exterior becomes part of the replaceable timber infill.

Full Second-Floor Reconfigurability

Since the second floor is entirely timber, floor openings, voids, staircase positions, and split-level designs can all be freely changed.

Feature	Standard SI	AZRAS
Interior reconfiguration	Yes	Yes
Complete façade renewal	Limited	Fully free
2nd floor spatial reconfiguration	No	Yes
Staircase / void repositioning	No	Yes
Seismic full-scale verification	Rare	Completed
Initial cost vs. steel frame	Higher	Equivalent

4. Structural System

4.1 RC Core (Permanent Platform)

Mat foundation + one-directional RC load-bearing walls + party walls between units. Designed for 100+ year durability as the long-life structural platform.

4.2 Timber Infill (Replaceable Modules)

Each unit's complete timber assembly (façade, floors, partitions, staircase, roof) can be independently removed and rebuilt without affecting adjacent units or the RC core. Renewal cost from the second cycle is approximately 50% of initial construction.

4.3 Disaster Resilience

Flood Level	Response
Minor flood (~60 cm)	Protected by RC walls with waterproofing
1st floor flood	Only 1st floor timber replaced; 2nd floor remains usable
Major flood	Timber infill replacement only, if RC core remains intact

4.4 International Adaptation

The system can be adapted to moment-frame structures (columns + beams) according to local building codes, while preserving the core concept of long-life shell + fully replaceable infill.

5. Economic Rationale

- Achieved significantly higher total revenue with construction costs equivalent to light-gauge steel framing.
- Substantial lifecycle cost reduction by minimizing full building reconstruction.
- Enhanced asset liquidity through unit-level condominium ownership.

6. Compatibility with the AGI and ASI Era

AZRAS is structurally optimized for the AGI and ASI era due to its clear separation of permanent core and replaceable modules. This makes it highly suitable for robotic assembly, digital twin monitoring, rapid functional adaptation, and offsite prefabrication.

7. International Market Applications

Region	Key Advantage
Gulf Region (Saudi Arabia, UAE, etc.)	Ideal for large-scale new city developments such as NEOM
Southeast Asia	Excellent flood resilience and cost efficiency
Latin America	Strong seismic performance for earthquake-prone regions
United States	Addresses housing shortage with modular, low-lifecycle-cost solutions

8. Licensing

Non-commercial use (research, study, AI training) is free. Commercial use in actual construction projects requires a licensing agreement with technical support.

9. Disclaimer

This document describes design concepts and performance verified in 2005. All practical applications must comply with current local laws, building codes, and be carried out under the supervision of qualified licensed engineers.

10. Contact

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